

Fabian Gittins — Publications

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PUBLICATION SUMMARY	Full list of publications can be found on Google Scholar , INSPIRE-HEP and NASA ADS . h-index —16 (according to Google Scholar) or 14 (according to INSPIRE-HEP). Both include collaboration papers. Top five cited —Excluding long-author-list papers. Citation counts according to Google Scholar. 1. Gittins, F., Andersson, N., Jones, D. I., <i>Modelling neutron star mountains</i>, Mon. Not. R. Astron. Soc. 500, 5570 (2021) [arXiv:2009.12794]. (82 citations) 2. Gittins, F., Andersson, N., <i>Modelling neutron star mountains in relativity</i>, Mon. Not. R. Astron. Soc. 507, 116 (2021) [arXiv:2105.06493]. (75 citations) 3. Gittins, F., Andersson, N., Pereira, J. P., <i>Tidal deformations of neutron stars with elastic crusts</i>, Phys. Rev. D 101, 103025 (2020) [arXiv:2003.05449]. (52 citations) 4. Pereira, J. P., Bejger, M., Andersson, N., Gittins, F., <i>Tidal deformations of hybrid stars with sharp phase transitions and elastic crusts</i>, Astrophys. J. 895, 28 (2020) [arXiv:2003.10781]. (44 citations) 5. Gittins, F., <i>Gravitational waves from neutron-star mountains</i>, Class. Quantum Gravity 41, 043001 (2024) [arXiv:2401.01670]. (37 citations)	
COLLABORATION PUBLICATIONS	Since 2026, I have been a co-author of refereed LIGO-Virgo-KAGRA and Einstein Telescope Collaboration publications. I list only non-collaboration articles below.	
SUBMITTED PUBLICATIONS	[28] Yu, H. et al., <i>Nonlinear hydrodynamics in spinning neutron stars: Theoretical universal relations and equilibrium solutions</i> [arXiv:2607.07943]. [27] Negri, L. et al., <i>Impact of the Einstein Telescope’s duty cycle on the estimation of binary black holes parameters</i> [arXiv:2606.19201].	
ACCEPTED PUBLICATIONS	[26] Gittins, F. et al., <i>Detecting tidal resonances in binary neutron stars</i> , Phys. Rev. Lett. 137, 011101 (2026) [arXiv:2606.06376].	
REFEREED PUBLICATIONS	Research articles [25] Andersson, N., Counsell, A. R., Gittins, F., Ghosh, S., <i>Tidal response of a relativistic star</i>, Phys. Rev. D 113, 064051 (2026) [arXiv:2511.05139]. [24] Yin, S., Andersson, N., Gittins, F., <i>A post-Newtonian approach to neutron star oscillations</i>, Class. Quantum Gravity 42, 235002 (2025) [arXiv:2504.06918]. [23] Pnigouras, P., Andersson, N., Gittins, F., Counsell, A. R., <i>Dynamical neutron star tides: the signature of a mode resonance</i>, Mon. Not. R. Astron. Soc. 542, 1375 (2025) [arXiv:2508.06416].	

- [22] Counsell, A. R., **Gittins, F.** et al., *Interface modes in inspiralling neutron stars: A gravitational-wave probe of first-order phase transitions*, **Phys. Rev. Lett.** **135**, 081402 (2025) [arXiv:2504.06181].
- [21] **Gittins, F.**, Andersson, N., Yin, S., *Perturbation theory for post-Newtonian neutron stars*, **Class. Quantum Gravity** **42**, 135014 (2025) [arXiv:2503.03345].
- [20] **Gittins, F.**, Andersson, N., *Neutron-star seismology with realistic, finite-temperature nuclear matter*, **Phys. Rev. D** **111**, 083024 (2025) [arXiv:2406.05177].
- [19] **Gittins, F.** et al., *Problematic systematics in neutron-star merger simulations*, **Phys. Rev. D** **111**, 023049 (2025) [arXiv:2409.13468].
- [18] Counsell, A. R., **Gittins, F.** et al., *Neutron star g modes in the relativistic Cowling approximation*, **Mon. Not. R. Astron. Soc.** **536**, 1967 (2025) [arXiv:2409.20178].
- [17] Counsell, A. R., **Gittins, F.**, Andersson, N., *The impact of nuclear reactions on the neutron-star g-mode spectrum*, **Mon. Not. R. Astron. Soc.** **531**, 1721 (2024) [arXiv:2310.13586].
- [16] Pnigouras, P., **Gittins, F.** et al., *The dynamical tides of spinning Newtonian stars*, **Mon. Not. R. Astron. Soc.** **527**, 8409 (2024) [arXiv:2205.07577].
- [15] Beri, A. et al., *AstroSat and NuSTAR observations of XTE J1739-285 during the 2019-2020 outburst*, **Mon. Not. R. Astron. Soc.** **521**, 5904 (2023) [arXiv:2303.13085].
- [14] **Gittins, F.** et al., *Modelling Neutron-Star Ocean Dynamics*, **Universe** **9**, 226 (2023) [arXiv:2304.05413].
- [13] **Gittins, F.**, Andersson, N., *The r-modes of slowly rotating, stratified neutron stars*, **Mon. Not. R. Astron. Soc.** **521**, 3043 (2023) [arXiv:2212.04892].
- [12] Andersson, N., **Gittins, F.**, *Formulating the r-mode Problem for Slowly Rotating Neutron Stars*, **Astrophys. J.** **945**, 139 (2023) [arXiv:2212.04837].
- [11] Andersson, N., **Gittins, F.** et al., *Building post-Newtonian neutron stars*, **Class. Quantum Gravity** **40**, 025016 (2023) [arXiv:2209.05871].
- [10] Riley, J. et al., *Rapid Stellar and Binary Population Synthesis with COMPAS*, **Astrophys. J. Suppl. Ser.** **258**, 34 (2022) [arXiv:2109.10352].
- [9] **Gittins, F.**, Andersson, N., *Modelling neutron star mountains in relativity*, **Mon. Not. R. Astron. Soc.** **507**, 116 (2021) [arXiv:2105.06493].
- [8] **Gittins, F.**, Andersson, N., Jones, D. I., *Modelling neutron star mountains*, **Mon. Not. R. Astron. Soc.** **500**, 5570 (2021) [arXiv:2009.12794].
- [7] **Gittins, F.**, Andersson, N., Pereira, J. P., *Tidal deformations of neutron stars with elastic crusts*, **Phys. Rev. D** **101**, 103025 (2020) [arXiv:2003.05449].
- [6] Pereira, J. P., Bejger, M., Andersson, N., **Gittins, F.**, *Tidal deformations of hybrid stars with sharp phase transitions and elastic crusts*, **Astrophys. J.** **895**, 28 (2020) [arXiv:2003.10781].
- [5] **Gittins, F.**, Andersson, N., *Population synthesis of accreting neutron stars emitting gravitational waves*, **Mon. Not. R. Astron. Soc.** **488**, 99 (2019) [arXiv:1811.00550].

Review articles

- [4] **Gittins, F.**, *Gravitational waves from neutron-star mountains*, **Class. Quantum Gravity** **41**, 043001 (2024) [arXiv:2401.01670].

Software articles

- [3] Riley, J. et al., *COMPAS: A rapid binary population synthesis suite*, *J. Open Source Softw.* **7**, 3838 (2022).

Conference proceedings

- [2] Thomas, A. Stevenson, E., **Gittins, F.** et al., *Galactic Archaeology with TESS: Prospects for Testing the Star Formation History in the Solar Neighbourhood*, *EPJ Web Conf.* **160**, 05006 (2017) [[arXiv:1610.08862](#)].

WHITE PAPERS

- [1] Dietrich, T. et al., *ESO Expanding Horizon White Paper: Revealing the properties of matter at supranuclear densities with gravitational waves* [[arXiv:2512.16971](#)].